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REVIEW
Educational Papers

The link between venous thromboembolism and atherothrombosis: epidemiology, pathophysiology, and implications for clinical care

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Venous thromboembolism (VTE), comprised of deep vein thrombosis (DVT) and pulmonary embolism (PE), remains a significant public health burden in the USA with an estimated 1 200 000 VTE cases in 2020. Atherothrombosis refers to thrombosis in the setting of atherosclerotic cardiovascular disease and includes type I myocardial infarction (MI), ischaemic stroke, and acute limb events with an estimated 720 000 MIs annually in the USA. Despite the historical view of VTE and atherothrombosis as unrelated entities recognized by 'red vs. white clots', these conditions are closely interconnected and share many pathophysiological features and clinical risk factors. This review provides a critical appraisal of the current evidence for the relationship between VTE and atherothrombosis, exploring potential mechanisms and risk factors linking the disorders, recent discoveries from basic and translational research, clinical investigation highlighting their commonalities, as well as the therapies that address both.

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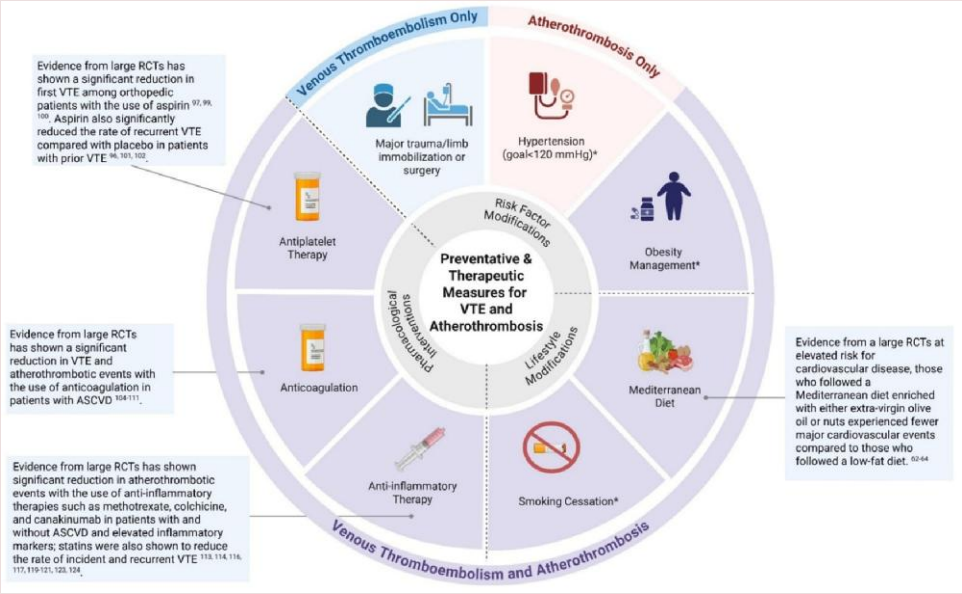
Graphical Abstract



Avenues for an integrated approach to prevention and treatment of atherothrombosis and venous thromboembolism. ASCVD, atherosclerotic cardiovascular disease; RCT, randomized controlled trial; VTE, venous thromboembolism. *No evidence from RCTs available.

Keywords Antiplatelet therapy • Atherothrombosis • Direct oral anticoagulants • Pulmonary embolism • Venous thromboembolism

Introduction

Atherothrombotic vascular disease, marked by potentially diffuse progressive arterial involvement, atherosclerotic plaque disruption, and accompanying thrombosis, is the primary cause of cardiovascular morbidity and mortality worldwide.[1,2] In 2019, the incidence rate of ischaemic stroke was 94 per 100 000 person-years in the USA, while recent data from Europe report an incidence rate of 90–100 per 100 000 person-years.[3] The incidence rate of acute limb ischaemia in 2020 was 5.9 per 100 000 person-years in the USA.[4,5] The American Heart Association reported the estimated incidence rate of acute MI to be 174.2 per 100 000 person-years and of recurrent MI cases to be 57.6 per 100 000 person-years in 2018.[6] Plaque rupture occurs when the fibrous cap separating the underlying atheroma from the circulating blood breaks, exposing the thrombogenic core of the plaque, as occurs in myocardial infarction (MI) and stroke.[7] Prevention and treatment of atherothrombosis typically involves blood pressure and glycaemic control, antiplatelet agents, and lipid lowering therapy.

Venous thromboembolism (VTE), comprising deep vein thrombosis (DVT) and pulmonary embolism (PE), is the third most common vascular disease after MI and stroke in the USA[8] with an estimated incidence rate of all VTE in the USA of 264.6 per 100 000 person-years in 2024. Data from several European countries report an annual VTE incidence of 150–200 cases per 100 000 person-years.[9–11] The pathophysiology of VTE is traditionally linked to the three domains of the Virchow's triad: endothelial injury, venous stasis, and hypercoagulability.[12] Based on the traditional thinking of venous thrombosis being less platelet based than arterial, anticoagulation has been the cornerstone for prevention and management of VTE.[13]

Although such conditions have been traditionally considered to be confined to arterial vs. venous sides of vasculature, pathophysiological, and clinical risk factors, and social determinants of health can similarly impact the development and outcomes of atherothrombosis and VTE and could, in part, explain the frequent coexistence of the two disorders.[12,14,15] This review examines current evidence on this relationship, delving into potential contributing mechanisms and risk factors to highlight the link between VTE and atherothrombosis that may ultimately inform more comprehensive strategies for prevention and management.

Epidemiology

Risk factors for atherothrombosis and venous thromboembolism

Chronic inflammatory or autoimmune disorders and infections significantly increase the risk of both atherothrombosis and VTE although the magnitude of risk may be different for both conditions (*Figure 1*).[16,17] For example, inflammatory bowel disease is associated with increased risk of both VTE [incidence rate ratio 2.36, 95% confidence interval (CI): 2.16–2.58] and atherothrombosis (incidence rate ratio 1.26, 95% CI: 1.11–1.44) likely driven by a hypercoagulable state secondary to inflammation and endothelial dysfunction.[18–20] Additionally, the risk of both VTE and atherothrombosis is increased among patients with rheumatoid arthritis (risk ratio 1.88, 95% CI: 1.65–2.15 for VTE[21]; risk ratio 1.48, 95% CI: 1.36–1.62 for incident cardiovascular disease[22]) compared with those without. Similarly, patients with thrombotic antiphospholipid syndrome are at higher risk of both arterial

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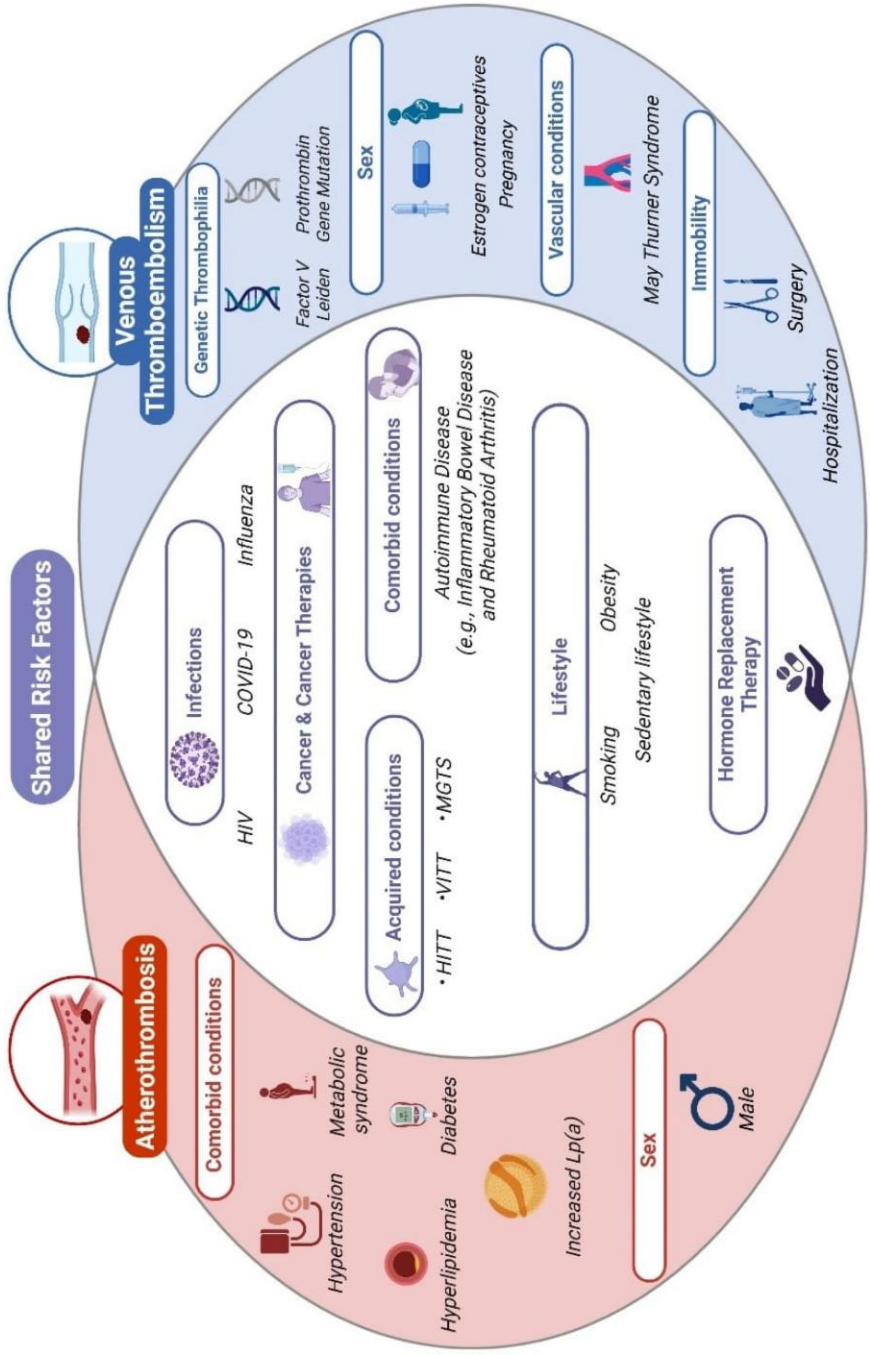

Figure 1 Shared risk factors for atherothrombosis and venous thromboembolism. ASCVD, atherosclerotic cardiovascular disease; CKD, chronic kidney disease; HITT, heparin-induced thrombosis and thrombocytopenia; HIV, human immunodeficiency virus; Lp(a), lipoprotein a; MGTS, monoclonal gammopathy of thrombotic significance; VITT, vaccine-induced thrombotic thrombocytopenia. *Studies on the association between elevated Lp(a) levels and VTE have yielded inconsistent results.

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and venous thromboses than individuals without antiphospholipid syndrome.[23–26] Influenza outbreaks and the recent COVID-19 pandemic highlight how infections can promulgate prothrombotic states, significantly increasing the risk of both venous and arterial thrombosis.[27–30] Likewise, human immunodeficiency virus infection increases the risk of both atherothrombosis (MI for example; risk ratio 1.79, 95% CI: 1.54–2.08[31]) and VTE (estimated 2–10 times increased risk[31]) through chronic inflammation, immune activation, endothelial dysfunction, and altered coagulation pathways, compared with non-infected patients.[32–34] Cancer and its therapies also increase the risk of both VTE and atherothrombosis: population-based studies revealed that patients with any type of cancer were at increased risk of VTE [hazard ratio (HR) 4.7, 95% CI: 4.5–4.9][35] and of atherothrombosis (HR 2.2, 95% CI: 2.1–2.3).[36] VTE and atherothrombotic events are the second leading cause of death in individuals with cancer.[35] The increased risk of thrombosis in cancer is multi-factorial, being driven by tumour-related hypercoagulable state, type of tumour (highest risks with brain and pancreatic cancers), patient-specific factors (comorbidities, social habits such as cigarette smoking, performance status), and treatment-related factors (such as surgery, radiation, chemotherapy, and catheter-associated thrombosis).[37]

Hormonal factors also have a role in the development of VTE and atherothrombosis. The use of oestrogen in oral contraceptives and hormone replacement therapy is associated with VTE risk and predisposition to atherothrombosis, particularly stroke, in women and transgender individuals.[38] Selective receptor oestrogen modulators such as tamoxifen, used for the treatment and prevention of breast cancer, are also associated with an increased risk of VTE (risk ratio 3.5, 95% CI: 2.1–6.0).[39] Pregnancy, associated with elevated oestrogen levels, also increases the risk of VTE. Data on the association between pregnancy and atherothrombosis are more nuanced. The presence of specific adverse pregnancy outcomes such as gestational hypertension, pre-eclampsia, and gestation diabetes mellitus appears to increase the risk of coronary atherosclerosis and stroke later in life.[40–43] It is unclear however whether uncomplicated pregnancies also carry such risk. The relationship between testosterone and the risk of VTE is supported by mixed evidence.[44] In a randomized trial of men with hypogonadism and pre-existing or at high risk of cardiovascular disease, testosterone-replacement therapy was not associated with significantly increased rate of major cardiovascular events compared with placebo (HR 0.96; 95% CI, 0.78–1.17).[45] Additionally, an additional study exhibited an association between current and recent testosterone therapy and increased short-term VTE risk, as compared with a non-use group.[46,47] Data on the association between masculinizing gender affirming hormone therapy and VTE remain limited and are direly needed.[48]

Genetic factors may also play a key role in thrombosis risk, such as Factor V Leiden for VTE, and lipoprotein(a) levels for atherothrombosis. Factor V Leiden and prothrombin gene mutations nearly double VTE risk.[49–51] More recently, the potential role of lipoprotein(a) in developing atherothrombosis and VTE has been explored. Elevated lipoprotein(a) levels are now a well-recognized risk factor for the development of atherothrombosis,[52] whereas clinical studies using Mendelian randomization have failed to show a consistent association between lipoprotein(a) and VTE.[53–57] Patients exposed to heparin can rarely develop heparin-induced thrombocytopenia, which presents as a drop in platelet count and increase the risk of atherothrombosis and VTE by platelet factor 4 antibodies production that activate platelets.[58] By a similar mechanism (production of platelet factor 4 antibodies), vaccine-induced thrombotic thrombocytopenia is rare condition that can cause atherothrombosis and VTE secondary to COVID-19 vaccination.[59]

Lifestyle risk factors, such as smoking status, a sedentary lifestyle, type of diet, and environmental stressors, are closely linked to atherothrombosis and VTE risk.[60–62] Smoking is a well-established risk factor for both VTE[63] and atherothrombosis.[64] Air pollution, particularly long-term exposure to particulate matter < 2.5 μm, significantly increases the risk of MI (relative risk 1.03, 95% CI: 1.02–1.04)[65] and shows a trend towards increased VTE risk (HR 1.07, 95% CI: 0.99–1.16).[66] Likewise, a Western diet has been linked to an increased risk of VTE and to the development of atherothrombotic disease.[67] Alternatively, the Mediterranean diet, rich in fatty fish, fruits, and vegetables, results in lower rates of atherothrombosis and VTE, compared with other diets.[68–70] Persistent stress and low occupational status in men are associated with an increased risk of pulmonary embolism.[71]

The above-described mutual risk factors for VTE and atherothrombosis help better explain the reciprocal relationship of increased risk of atherothrombosis in patients with VTE and the increased risk of VTE in patients with atherothrombosis. Early studies hinted at a potential link between atherothrombosis and VTE, noting that the presence of carotid artery plaques nearly doubled the risk of VTE.[12,14,15,72] Additionally, patients with known peripheral arterial disease and severely low ankle-brachial index were associated with an increased risk of subsequent VTE (HR 1.46, 95% CI: 1.31–1.64).[73] Increased burden of atherosclerosis, manifested by a higher number of symptomatic vascular territories, was found to also be associated with a graded increase in the rates of VTE, with those showing evidence of coronary, cerebrovascular, and peripheral arterial atherosclerosis bearing the greatest risk.[74] Separately, patients with a new VTE diagnosis were found to be at an elevated risk of subsequent stroke and MI, particularly within 1-year post-VTE.[75]One of the explanations behind the increased risk of VTE in patients with atherothrombosis can be a complication of atherothrombosis care such as hospitalization, surgery, or procedure and vice-versa. Rarely, atherothrombotic events can also occur as a direct consequence of VTE in patients with patent foramen oval in the form of paradoxical systemic embolism.

While cardiometabolic risk factors, including hypertension, diabetes mellitus, dyslipidaemia, and metabolic syndrome, are known to be causally associated with atherothrombosis, their mechanistic role in VTE is elusive.[64] Genetically elevated low-density lipoprotein cholesterol levels have been linked to an increased risk for VTE compared with patients with genetically normal cholesterol levels.[76] Diabetes, hypertension, and metabolic syndrome, particularly in males, have also been associated with an increased risk of VTE compared with individuals without these conditions.[61,77–80] However, whether these factors act as true independent risk factors for VTE remains unclear.

Pathophysiology of atherothrombosis and venous thromboembolism: Virchow's triad and the role of inflammation

The concept of Virchow's triad of endothelial injury, hypercoagulability, and stasis of blood flow has long provided a framework for venous thrombosis and atherothrombosis formation (*Figure 2*). Reduced blood flow and stasis can lead to the aggregation of procoagulant factors in the blood, which can cause an imbalance in pro- and anticoagulant pathways and lead to thrombus formation.[67,81–83] Accordingly, higher rates of VTE are thus seen in hospitalized patients, those who have undergone surgery, long-haul travel, pregnancy and patients with anatomical common left common iliac vein compression, known as May–Thurner Syndrome. On the other hand, stasis does not appear play a significant role in the pathogenesis of atherothrombosis.[81]

Hypercoagulability also plays a crucial role in thrombus formation. A hypercoagulable state can be driven by inherited thrombophilias (factor V Leiden, prothrombin gene mutation) or acquired conditions (antiphospholipid syndrome, chronic inflammatory diseases, and infections). For example, antiphospholipid antibodies confer an increased risk of both atherothrombosis and VTE.[84–86] Endothelial injury has also been well-established in the pathogenesis of thrombosis and can be

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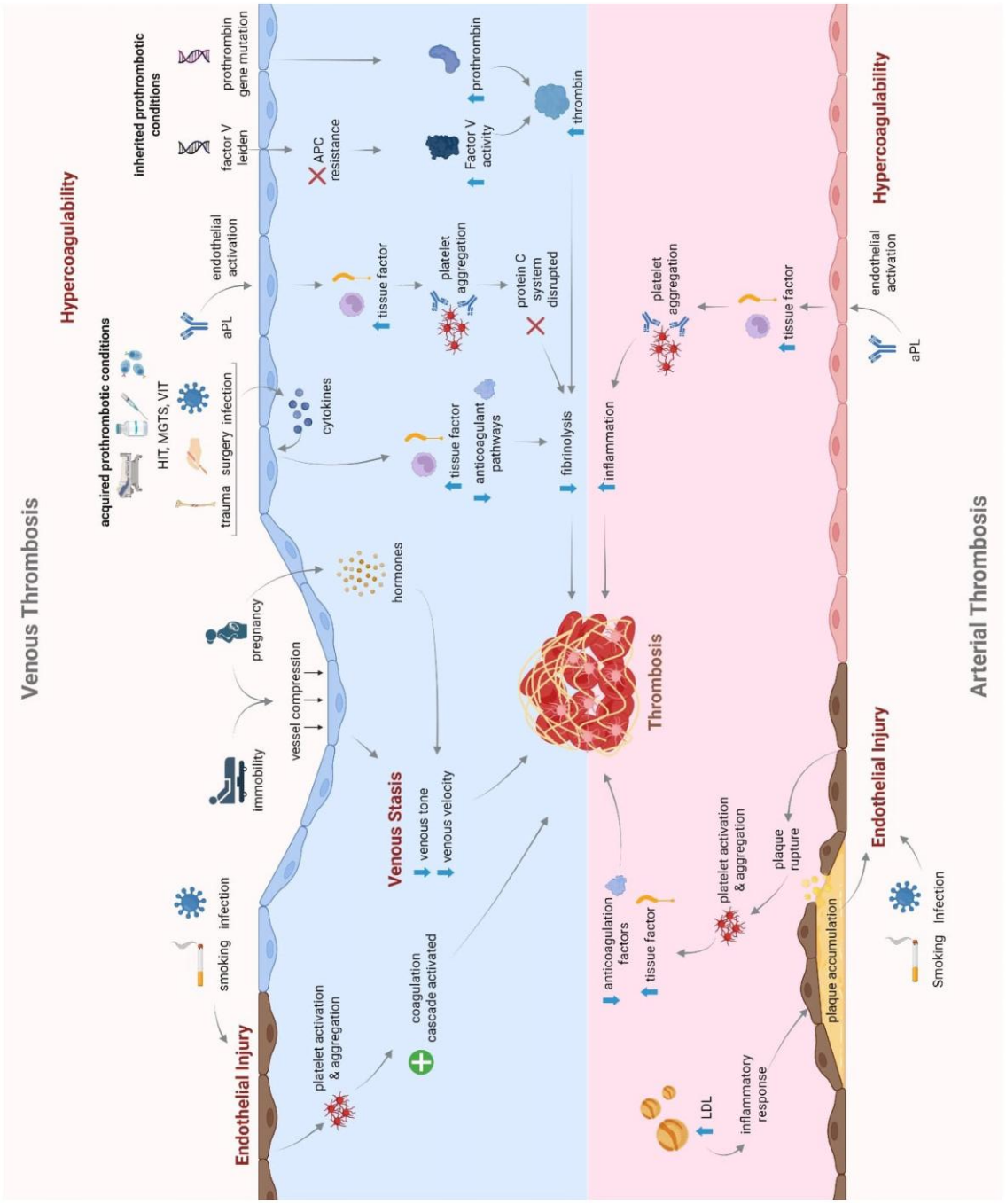

Figure 2 Pathophysiological mechanisms of atherothrombosis and venous thromboembolism. APC, activated protein C; aPL, antiphospholipid; HITT, heparin-induced thrombosis and thrombocytopenia; LDL, low-density lipoprotein; MGTS, monoclonal gammopathy of thrombotic significance; VITT, vaccine-induced thrombotic thrombocytopenia.

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traumatic—due to direct damage of the blood vessel wall—or secondary to inflammation.

While often viewed as distinct processes, inflammation and coagulation are tightly linked mechanisms by which the body responds to injury and infection.[87] Immunothrombosis is the process by which the innate immune system promotes localized thrombus formation within the vasculature as a host defence mechanism, in response to infection or tissue injury.[88] Thromboinflammation refers to the pathologic state in which the interplay between thrombosis and inflammation become dysregulated, resulting in excessive or systemic clot formation and tissue injury which occurs in infections, autoimmune diseases and cardiovascular disease.[88] Tissue factor, a type I integral membrane glycoprotein located in the sub-endothelial tissues, connects the inflammatory and coagulation pathways. Tissue factor is up-regulated in response to infection, oxygen-free radicals, inflammatory cytokines, physical and/or chemical damage. In turn, tissue factor and factor VIIa form a complex, activate coagulation factors IX and X, and ultimately set the coagulation cascade in motion.[89] Additionally, neutrophil extracellular traps (NETs), which are web-like structures composed of decondensed chromatin (DNA and histones) are released by neutrophils in response to various stimuli, including pathogens, activated platelets, and inflammatory pathogens through a process called NETosis.[90] These NETs provide a scaffold that promotes platelet adhesion, activation, and aggregation. Dysregularted NETs formation and increased interleukins play a key role in the pathogenesis of VTE and atherothrombosis, as both are triggered by enhanced inflammation and clot formation.[91–94] This close relationship between inflammation and coagulation helps explain the elevated risk of atherothrombosis and VTE in chronic inflammatory and autoimmune diseases.

Implications of the link between atherothrombosis and VTE on clinical care

Lifestyle modification

Lifestyle modification, including diet, exercise, and smoking cessation, is believed to be paramount to primary and secondary prevention of atherothrombosis and to potentially play a role in altering VTE risk. Although clinical outcome trials do not exist to inform on the impact of the type and duration of exercise to reduce incident atherothrombotic cardiovascular diseases, it is well known that aerobic exercise can improve some of its key risk factors, including dyslipidaemia, hypertension, glucose metabolism, and insulin resistance.[95] Separately, in patients with established coronary artery disease, exercise-based cardiac rehabilitation has been shown to decrease the risk of MI, but not mortality compared with no cardiac rehabilitation.[96] With regards to VTE, prospective cohort studies suggest that regular physical activity may be associated with a reduced rate of incident VTE, likely driven by decreased sedentary lifestyle.[97] Smoking cessation is well-accepted strategy to reduce the risk of atherothrombosis and VTE. In a meta-analysis of prospective cohort studies of patients with coronary artery disease, results suggest that smoking cessation is associated with a decreased rate of all-cause mortality compared with patients who continue smoking (risk ratio 0.64, 95% CI: 0.58–0.71).[98] Prospective cohort studies suggest that greater fish, fruit, and vegetables consumption is associated with a decreased incidence of VTE compared with other diets.[68] In an RCT of 7447 patients who were at high risk but without prior cardiovascular disease, subjects assigned to a Mediterranean diet had significantly lower risk of a first cardiovascular event (MI, stroke, or cardiovascular death) compared with patients in the control group (HR 0.69, 95% CI: 0.53–0.91, number needed to treat was 67).[99] In another RCT of 1002 patients with established coronary heart disease, patients assigned to a Mediterranean diet had a lower risk of a composite of MI, stroke, or cardiovascular death compared with patients assigned to a low-fat diet (HR 0.73, 95% CI: 0.55–0.97, number needed to treat was 21).[100] Neither of these trials have reported results for VTE.

The influenza vaccine has been associated with a decreased risk of atherothrombosis. In a RCT of 2571 patients with recent MI or high-risk stable coronary artery disease, patients assigned to receive an inactivated influenza vaccine had lower rates of a composite of all-cause death, MI, or stent thrombosis compared with placebo (HR 0.77, 95% CI: 0.52–0.99), stressing the role of influenza vaccination in high-risk cardiovascular patients.[101] Trials did not report VTE outcomes in patients with influenza vaccination; however, emerging evidence reveal that vaccination is likely to be associated with decreased rates of VTE.[102]

Pharmacological interventions

Pharmacotherapy for preventing and treating VTE and atherothrombosis involves managing risk factors, as well as anticoagulation, antiplatelet therapy, or both (Graphical Abstract). RCTs were considered for inclusion in the tables only if the tested treatment was assessed for effect both on atherothrombosis and on VTE. Trials that did not assess either or both, such as those for stroke prevention in atrial fibrillation, are not discussed here.

Antiplatelet and anticoagulant therapy

Antithrombotic therapy has long been recognized as playing a pivotal role in the prevention and management of atherothrombosis and VTE. Traditionally, antiplatelet agents have been considered for the former and anticoagulants for the latter, even though intravenous heparin is essential in the acute management of atherothrombotic events. More recent data suggest this distinction may not be as well-defined.

Multiple prior randomized trials have shown the effect of aspirin on reducing atherothrombotic events.[103] Antiplatelet agents have been shown to be effective in the primary prevention of VTE[104] with multiple RCTs suggesting comparable effectiveness vs. parenteral anticoagulation in patients undergoing orthopaedic surgery,[104–106] and have shown modest benefit in the secondary prevention of VTE (*Table 1*).

In an RCT of 403 patients with unprovoked VTE who completed 6–18 months of anticoagulation, aspirin significantly reduced the risk of recurrent VTE compared with placebo alone (HR 0.58, 95% CI: 0.36–0.93).[108] Alternatively, in another RCT of 822 patients with unprovoked VTE who completed 6 weeks—24 months of initial anticoagulation, aspirin did not significantly reduce the rate of recurrent VTE compared with placebo alone (HR 0.74, 95% CI: 0.52–1.05). However, the rate of major vascular events was significantly lower among patients assigned to aspirin (HR 0.66, 95% CI: 0.48–0.92).[107] In a meta-analysis of the previous two trials, aspirin significantly reduced the rate of recurrent VTE (HR 0.68, 95% CI: 0.51–0.90).[109] Finally, in a pre-specified analysis of two trials that enrolled 47 611 patients with a history of MI, stroke, or peripheral artery disease, dual antiplatelet therapy (aspirin or clopidogrel plus ticagrelor or vorapaxar) significantly reduced the rate of VTE (HR 0.71, 95% CI: 0.56–0.89) compared with single antiplatelet therapy.

DOACs have now become standard of care for stroke prevention in atrial fibrillation[110] and management of acute VTE.[111–114] Recent studies have also highlighted the use of anticoagulants for the management of atherothrombosis (*Table 2*).

In a trial of 27 395 patients with stable atherosclerotic cardiovascular disease, rivaroxaban 2.5 mg twice daily plus antiplatelet therapy significantly reduced the rate of a composite of cardiovascular death, stroke, or MI compared with aspirin alone (HR 0.76, 95% CI: 0.66–0.86)[117] and the rate of VTE compared with aspirin alone (HR 0.61, 95% CI: 0.37–

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Table 1 Anti-platelet therapies for the management of atherothrombosis and venous thromboembolism

Author, year	Study design	Treatment vs. control	Study population	Results summary
A. Atherothrombosis				
ATT, 2009[103]	Meta-analysis of RCTs	Any low-dose aspirin vs. Placebo	95 000 patients in 6 RCTs, 17 000 patients in 17 RCTs	Aspirin significantly reduced the rate of a first serious vascular event for the primary and secondary cardiovascular interventions. Aspirin significantly increased the rate of a major extracranial bleed (RR 1.54, 95% CI: 1.30–1.82)
B. VTE				
PEP trial, 2000[106]	RCT	Aspirin 81 mg vs. Placebo	13 556 undergoing surgeries for hip fracture. 4088 undergoing either elective hip or knee arthroplasty	Thromboprophylaxis with aspirin reduced the rates of the VTE. (HR 0.71, 95% CI: 0.54–0.94). Additionally, thromboprophylaxis with aspirin did not significantly reduce rates of VTE (HR 0.83, 95% CI: 0.47–1.47).
CRISTAL Study, 2022[105]	RCT	Aspirin 100 mg vs. Enoxaparin 40 mg	9711 patients undergoing hip or knee arthroplasty	Enoxaparin was superior for VTE prevention compared with aspirin as indicated by, fewer calf DVTs in enoxaparin group (risk difference 1.97%, 95% CI: 0.54–3.41%).
PREVENT CLOT METRC, 2023[104]	RCT	Aspirin 81 mg twice daily vs. Enoxaparin 30 mg twice daily	12 211 patients undergoing extremity fracture	Aspirin 81 mg twice daily was non-inferior to enoxaparin 30 mg twice daily for the prevention of death (risk difference 0.05%, 95% CI: −0.27–0.38%). Aspirin arm had significantly higher rates of DVT compared with enoxaparin (risk difference 0.80%, 95% CI: 0.28–1.31%).
Brighton *et al.*, 2012[107]	RCT (ASPIRE)	Aspirin 100 mg vs. Placebo	822 patients with unprovoked VTE who completed 6 weeks —24 months of initial anticoagulation	Aspirin did not significantly reduce the rate of recurrent VTE compared with placebo alone (HR 0.74, 95% CI: 0.52–1.05). The rate of major vascular events[a] was significantly lower among patients assigned to aspirin (HR 0.66, 95% CI: 0.48–0.92).
Becattini *et al.*, 2012[108]	RCT (WARFASA)	Aspirin 100 mg vs. Placebo	403 patients with unprovoked VTE who completed 6–18 months of anticoagulation	Aspirin significantly reduced the risk of recurrent VTE compared with placebo (HR 0.58, 95% CI: 0.36–0.93). Aspirin significantly reduced the rate of recurrent VTE (HR 0.68, 95% CI: 0.51–0.90).[109]
Cavallari *et al.*, 2017[74]	Pre-specified analysis of RCTs TRA2P-TIMI 50 and PEGASUS-TIMI 54	Aspirin or clopidogrel + ticagrelor or vorapaxar vs. SAPT	47 611 patients with a history of MI, stroke, or peripheral artery disease	Dual antiplatelet therapy (aspirin or clopidogrel + ticagrelor or vorapaxar) significantly reduced the rate of VTE (HR 0.71, 95% CI: 0.56–0.89) compared with SAPT.

ATT, antithrombotic trialists; CI, confidence interval; DVT, deep vein thrombosis; HR, hazard ratio; MI, myocardial infarction; PEP, pulmonary embolism prevention; RCT, randomized controlled trial; RR, risk rate; SAPT, single antiplatelet therapy (aspirin or clopidogrel); VTE, venous thromboembolism.
[a]Major vascular events included a composite of VTE, MI, stroke, or cardiovascular death.

1.00).[120] In another RCT of 6564 patients with peripheral artery disease who have undergone lower-extremity revascularization, rivaroxaban 2.5 mg twice daily plus aspirin significantly reduced the rate of a composite of acute limb ischaemia, MI, stroke, or cardiovascular death compared with aspirin alone (HR 0.85, 95% CI: 0.76–0.95).[118] The addition of DOACs to clopidogrel in patients with stable atherosclerotic cardiovascular disease has not been explored yet. DOACs have also been investigated in patients with recent acute coronary syndrome. In an RCT of 15 526 patients with recent acute coronary syndrome, rivaroxaban 2.5 or 5 mg twice daily significantly reduced the rate of a composite of cardiovascular death, MI, or stroke compared with placebo (HR 0.84, 95% CI: 0.74–0.96) with increased rate of major bleeding in the rivaroxaban arm.[121] Separately, in another RCT of 7392 patients with acute coronary syndrome, apixaban 5 mg twice daily did not significantly reduce the rate of a composite of cardiovascular death, MI, or stroke compared with placebo (HR 0.95, 95% CI: 0.80–1.11) and major bleeding was significantly higher in the apixaban arm (HR 2.59, 95% CI: 1.50–4.46).[115] Finally, in an RCT of 172 patients with post-thrombotic syndrome, who underwent venous stent placement of the iliofemoral veins or inferior vena cava, there was no

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Table 2 Anticoagulation therapies for the management of atherothrombosis and venous thromboembolism

Author, year	Study design	Treatment vs. control	Study population	Results summary
A. Atherothrombosis				
Alexander *et al.*, 2011[115]	RCT (APPRAISE-2)	Apixaban 5 mg twice daily vs. placebo	7392 patients with recent acute coronary syndrome	Apixaban did not significantly reduce the rate of a composite of cardiovascular death, MI, or stroke compared with placebo (HR 0.95, 95% CI: 0.80–1.11) and major bleeding was significantly higher among the apixaban arm (HR 2.59, 95% CI: 1.50–4.46).
Mega *et al.*, 2012[116]	RCT (ATLAS ACS 2-TIMI 51)	Rivaroxaban 2.5 mg or 5 mg twice daily vs. placebo	15 526 patients with recent acute coronary syndrome	Rivaroxaban significantly reduced the rate of a composite of cardiovascular death, MI, or stroke compared with placebo (HR 0.84, 95% CI: 0.74–0.96) with increased rate of major bleeding in the rivaroxaban arm.
Eikelboom *et al.*, 2017[117]	RCT (COMPASS)	Rivaroxaban 2.5 mg twice daily + aspirin 81 mg daily vs. Aspirin 81 mg daily	23 396 patients with stable ASCVD	Rivaroxaban 2.5 mg twice daily + aspirin significantly reduced the rate of a composite of cardiovascular death, stroke, or MI compared with aspirin alone (HR 0.76, 95% CI: 0.66–0.86).
Bonaca *et al.*, 2020[118]	RCT (VOYAGER PAD)	Rivaroxaban 2.5 mg twice daily + aspirin 81 mg daily vs. Aspirin 81 mg daily	6564 patients with PAD who have undergone lower-extremity revascularization	Rivaroxaban 2.5 mg twice daily + aspirin significantly reduced the rate of a composite of acute limb ischaemia, MI, stroke, or cardiovascular death compared with aspirin alone (HR 0.85, 95% CI: 0.76–0.95).
Barco *et al.*, 2025[119]	RCT (ARIVA)	Rivaroxaban 20 mg daily + aspirin 100 mg daily vs. rivaroxaban 20 mg daily	172 patients with post-thrombotic syndrome who underwent venous stenting of iliofemoral veins or IVC	There was no significant difference in the overall primary patency rate at 6 months between rivaroxaban 20 mg alone vs. rivaroxaban and aspirin 81 mg (absolute risk difference 2.4%, 95% CI: −13.6 to 18.0).
B. VTE				
Pogosova *et al.*, 2022[120]	RCT (COMPASS)	Rivaroxaban 2.5 mg twice daily + aspirin 81 mg daily vs. Aspirin 81 mg daily	23 396 patients with stable ASCVD	Rivaroxaban 2.5 mg twice daily + aspirin reduced the rate of VTE compared with aspirin alone (HR 0.61, 95% CI: 0.37–1.00).

AF, atrial fibrillation; ASCVD, atherosclerotic vascular disease; CI, confidence interval; DOACs, direct oral anticoagulants; HR, hazard ratio; IVC, inferior vena cava; MI, myocardial infarction; PAD, peripheral artery disease; RCT, randomized controlled trial; VTE, venous thromboembolism.

Table 3 Randomized controlled trials of inflammatory and lipid-lowering therapies for prevention or treatment of atherothrombosis and venous thromboembolism

Author, year	Trial name	Treatment vs. control	Study population	Results summary
A. Atherothrombosis				
Ridker *et al.*, 2008[123]	JUPITER	Rosuvastatin 20 mg vs. Placebo	17 802 healthy patients without hyperlipidaemia but with elevated hs-CRP levels	Rosuvastatin was associated with significantly reduced rates of a composite of stroke, MI, unstable angina, or cardiovascular death (HR 0.56, 95% CI: 0.46–0.69) compared with placebo.
Yusuf *et al.*, 2016[124]	HOPE-3	Rosuvastatin 10 mg vs. Placebo	12 705 patients without known cardiovascular disease but at intermediate risk	Rosuvastatin was associated with significantly lower rates of a composite of MI, stroke, or cardiovascular death compared with placebo (HR 0.76, 95% CI: 0.64–0.91).

Continued

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Table 3 ***Continued***

Author, year	Trial name	Treatment vs. control	Study population	Results summary
Ridker *et al.*, 2017[125]	CANTOS	Canakinumab (50 mg, 150 mg, 300 mg) vs. Placebo	10 061 patients with previous MI and hs-CRP ≥2 mg/L	Canakinumab 150 mg every 3 months significantly reduced the rate of a composite of MI, stroke, or cardiovascular death compared with placebo (HR 0.85, 95% CI: 0.74–0.98).
Ridker *et al.*, 2019[126]	CIRT	Methotrexate (15 to 20 mg weekly) vs. Placebo	4786 patients with stable ASCVD	Low-dose methotrexate was not associated with a significant reduction in the rates of a composite of MI, stroke, or cardiovascular death, compared with placebo (HR 0.96, 95% CI: 0.79–1.16)
Tardif *et al.*, 2019[127]	COLCOT	Colchicine 0.5 mg vs. Placebo	4745 patients with recent MI	Colchicine was associated with a significantly lower rate of a composite of cardiovascular death, MI, stroke, or angina requiring coronary revascularization, compared with placebo (HR 0.77, 95% CI: 0.61–0.96).
Nidorf *et al.*, 2020[128]	LoDoCo2	Colchicine 0.5 mg vs. Placebo	5522 patients with chronic coronary disease	Colchicine was associated with a significantly decreased rate of composite of cardiovascular death, MI, or stroke, compared with placebo (HR 0.69, 95% CI: 0.57–0.83)
Jolly *et al.*, 2024[129]	CLEAR	Colchicine 0.5 mg vs. Placebo	7062 patients with recent MI	Colchicine did not significantly reduce the rate of a composite of death from cardiovascular cause, recurrent MI, or stroke compared with placebo (HR 0.99, 95% CI: 0.85–1.16).
B. VTE				
Glynn, *et al.*, 2009[130]	JUPITER	Rosuvastatin 20 mg vs. Placebo	17 802 healthy patients without hyperlipidaemia but with elevated hs-CRP levels	Rosuvastatin was associated with significantly reduced rates of a composite of first VTE compared with placebo (HR 0.57, 95% CI: 0.37–0.8).
Marston *et al.*, 2020[131]	Meta-analysis	PCSK9i + statins vs. statins alone	46 488 patients in 2 RCTs	PCSK9i + high-intensity statins significantly reduced the risk of incident VTE (HR 0.69, 95% CI: 0.53–0.90) compared with high-intensity statin only.
Delluc *et al.*, 2021[132]	SAVER	Rosuvastatin 20 mg vs. Placebo	312 patients with VTE	Rosuvastatin plus anticoagulation compared with anticoagulation alone showed 3 recurrent VTE events in the rosuvastatin group and 2 in the control group. A larger trial is ongoing, and results are awaited (NCT04319627).
Wang *et al.*, 2024[133]	Systematic review and meta-analysis[a]	Colchicine 0.5 mg vs. placebo	Patients with various conditions prescribed colchicine.	In a systematic review and meta-analysis of various colchicine RCTs that reported VTE data, colchicine was not associated with decreased VTE rates compared with placebo (risk ratio 1.22, 95% CI: 0.86–1.74).
Farmakis *et al.*, 2024[134]	Systematic review and meta-analysis[a]	Different lipid-lowering therapies[b] vs. placebo	254 933 patients in 45 RCTs of statins, ezetimibe, or PSCK9i that reported VTE outcomes	PSCK9i + high-intensity statin was associated with the largest reduction in VTE risk (RR 0.59, 95% CI: 0.43–0.80), while there was a trend towards reduced VTE risk for high-intensity (RR 0.84, 95% CI: 0.70–1.02) and low-/moderate-intensity (0.89, 95% CI: 0.79–1.00) statin monotherapy compared with placebo. When compared with low-/moderate-intensity statin monotherapy, PCSK9i + statins significantly reduced the risk of VTE (RR 0.66, 95% CI: 0.49–0.89). Ezetimibe monotherapy did not affect VTE risk compared with placebo.

CI, confidence interval; HR, hazard ratio; hs-CRP, high-sensitivity C-reactive protein; MI, myocardial infarction; PCSK9i, proprotein convertase subtilisin/kexin type 9 inhibitors; RCT, randomized controlled trial; RR, risk ratio; VTE, venous thromboembolism.
[a]Not a clinical trial.
[b]Lipid-lowering therapy regimens included low/moderate/high intensity statins, ezetimibe, PCSK9i, or combination therapies.

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significant difference in the overall primary patency rate at 6 months between rivaroxaban 20 mg alone vs. rivaroxaban and aspirin 100 mg (absolute risk difference 2.4%, 95% CI: −13.6 to 18.0), suggesting that rivaroxaban alone is sufficient to maintain stent patency.[119]

This review did not address the question of what should be done in patients receiving antiplatelet therapy for atherosclerotic cardiovascular disease who develop a venous thromboembolic event requiring anticoagulation as the scope of this review is limited to exploring where antiplatelets are used for VTE and anticoagulants for atherothrombosis.

Anti-inflammatory and lipid-lowering therapies

Recent clinical trials have explored targeting inflammation to reduce thrombotic events (*Table 3*).[122]

In a RCT of 10 061 patients with prior MI and elevated levels of high-sensitivity C-reactive protein, canakinumab, an anti-interleukin-1$_\beta$ monoclonal antibody, was associated with decreased rates of recurrent MI and coronary revascularization (HR 0.88, 95% CI: 0.79–0.97), independent of lipid-level lowering, although this is limited by the risk of fatal infections.[125] VTE outcomes were not reported in the trial. Trials on colchicine for patients with recent MI have shown more conflicting results. In a RCT of 4745 patients with recent MI, colchicine was associated with a significantly lower rate of a composite of cardiovascular death, MI, stroke, or angina requiring coronary revascularization, compared with placebo (HR 0.77, 95% CI: 0.61–0.96).[127] However, in another RCT of 7062 patients with recent MI, colchicine did not significantly reduce the rate of a composite of death from cardiovascular cause, recurrent MI, or stroke compared with placebo (HR 0.99, 95% CI: 0.85–1.16).[135] In a RCT of 5522 patients with chronic coronary disease, colchicine was associated with a significantly decreased rate of composite of cardiovascular death, MI, or stroke, compared with placebo (HR 0.69, 95% CI: 0.57–0.83).[128] Randomized data on colchicine and VTE are more limited. In a systematic review and meta-analysis of various colchicine RCTs that reported VTE data, colchicine was not associated with decreased VTE rates compared with placebo (Risk Ratio 1.22, 95% CI: 0.86–1.74).[133] Finally, in an RCT of 4786 patients with stable atherosclerotic cardiovascular disease, low-dose methotrexate was not associated with a significant reduction in the rates of a composite of MI, stroke, or cardiovascular death, compared with placebo (HR 0.96, 95% CI: 0.79–1.16).[126]

While the role of statin therapy for reducing atherothrombotic events is well established,[136] their role in VTE and primary prevention of atherothrombosis is emerging. In a trial of healthy patients without hyperlipidaemia but with elevated high-sensitivity C-reactive protein levels, rosuvastatin was associated with significantly reduced rates of a composite of stroke, MI, unstable angina, or cardiovascular death (HR 0.56, 95% CI: 0.46–0.69), and a composite of first VTE compared with placebo (HR 0.57, 95% CI: 0.37–0.8).[123,130] Separately, in another RCT of 12 705 patients without known cardiovascular disease but at intermediate risk, rosuvastatin was associated with significantly lower rates of a composite of MI, stroke, or cardiovascular death compared with placebo (HR 0.76, 95% CI: 0.64–0.91).[124] In an individual patient data meta-analysis of the prior two mentioned RCTs, rosuvastatin was associated with decreased rates of first VTE compared with placebo (HR 0.53, 95% CI: 0.37–0.75).[137] An ongoing clinical trial (SAVER, Statins for Venous Event Reduction in Patients with Venous Thromboembolism, NCT04319627) is comparing rosuvastatin with a direct oral anticoagulant compared with a direct oral anticoagulant only for the prevention of recurrent VTE in patients with prior VTE. A pilot trial of 312 patients with VTE randomized to rosuvastatin plus anticoagulation compared with anticoagulation alone observed 3 recurrent VTE events in the rosuvastatin group and 2 in the control group.[132] Other lipid-lowering therapies including ezetimibe and proprotein convertase subtilisin/kexin type 9 inhibitors (PCSK9i) in combination with statins were also shown to reduce the risk of atherothrombosis[138–140]; to date however, there is no trial evaluating the effect of ezetimibe or proprotein convertase subtilisin/kexin type 9 inhibitors for the prevention of venous thromboembolism. In a meta-analysis of the two PCSK9i RCTs for atherothrombosis reduction, the combination of high-intensity statin with PCSK9i significantly reduced the risk of incident VTE (HR 0.69, 95% CI: 0.53–0.90) compared with high-intensity statin only, a reduction possibly mediated by a reduction in Lp(a) levels as there was a significant interaction between baseline Lp(a) levels and magnitude of VTE risk reduction.[131] In a separate systematic review and meta-analysis of 45 RCTs of statins (high- or low-/moderate-intensity), ezetimibe, or PSCK9i, the combination of PSCK9i with high-intensity statin was associated with the largest reduction in VTE risk (RR 0.59, 95% CI: 0.43–0.80), while there was a trend towards reduced VTE risk for high-intensity (RR 0.84, 95% CI: 0.70–1.02) and low-/moderate-intensity (0.89, 95% CI: 0.79–1.00) statin monotherapy compared with placebo.[134] When compared with low-/moderate-intensity statin monotherapy, the combination of PCSK9i and statins significantly reduced the risk of VTE (RR 0.66, 95% CI: 0.49–0.89).[134] Additionally, ezetimibe monotherapy did not affect VTE risk compared with placebo.[141] The results of these studies shed further light on the existence of crossing pathways between atherothrombosis and VTE.

Conclusion

While atherothrombosis and VTE may involve some distinct pathways, risk factors, and treatment approaches, an emerging body of evidence indicates significant overlap in their pathogenesis, risk factors, and prevention and treatment strategies. This review highlights such commonalities, including shared risk factors, underlying mechanisms, and pharmacologic interventions. Further research is needed to fully elucidate the interplay between VTE and atherothrombosis and to guide more effective preventive and management strategies.

Author contributions

Antoine Bejjani (Conceptualization, Methodology, Writing—original draft, Writing—review & editing), Candrika D. Khairani (Writing—original draft, Visualization), Darsiya Krishnathasan (Writing—original draft, Visualization), Samantha M. Rizzo (Writing—original draft, Visualization), Alyssa Sato (Writing—original draft, Visualization), Bassil Bacare (Writing—original draft, Visualization), Giovanni Scimeca (Writing—original draft, Visualization), Anthony Tristani (Writing—original draft, Visualization), Andre Armero (Writing—original draft, Visualization), Nicole Porio (Writing—original draft, Visualization), Behnood Bikdeli (Conceptualization, Methodology, Writing—original draft, Writing—review & editing), Gregory Piazza (Conceptualization, Methodology, Writing—original draft, Writing—review & editing).

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Data availability

Data in this manuscript is publicly available.

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